

# Yoho: Enabling a seamless online-to- offline shopping experience with a scalable database

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ABOUT YOH



By migrating to C  
Yoho boosts its a  
stay competitive  
powered dynamic  
and improve its r  
warehouse efficie  
innovative data-d  
applications.

## About Yoho

Founded in 2013, electronics products online shop Yoho is Hong Kong's first online-to-offline (O2O) ecommerce company, which means customers can purchase items through its website or in its dedicated stores. With more than 1.2 million monthly website visitors, Yoho tops the ranking of Hong Kong traffic and was ranked number eight on the 2019 Deloitte Hong Kong Tech Fast awards based on its revenue growth of 167% between 2017 to 2019.

Industries: Retail & Consumer Goods

## Google Cloud results

- Modernized backend database to handle 5X increase in traffic spikes during COVID-19
- Increases store and warehouse efficiencies by integrating point-of-sale and inventory management on Cloud SQL
- Hosts Yoho's proprietary AI bot on Compute Engine to attract customers with competitive pricing

Location: Hong Kong

Improve shopping experience 1.2M customers

(<https://cloud.google.com/sql/>), [Compute Engine](#)

Google Cloud (<https://cloud.google.com/>) [Cloud Translation API](#)

Cloud IAM (<https://cloud.google.com/iam/>) [Contact Center AI](#)

Cloud SQL (<https://cloud.google.com/sql/>) [Contact Center AI](#)

Compute Engine

(<https://cloud.google.com/compute/>)

[Enterprise Edition](#)

Cloud Translation API

(<https://cloud.google.com/translate/>)

Contact Center AI

(<https://cloud.google.com/solutions/contact-center/>)

Dialogflow Enterprise Edition

(<https://cloud.google.com/dialogflow-enterprise/>)

Did you know that 98% of Hong Kong shoppers

research products online

(<https://www.wirecard.com/uploads/knowledge-hub/white-paper/wirecard-global-consumer-survey.pdf>)

before buying them in stores? Searching for bargains online before visiting the physical store to test the actual products have become a norm, especially in a country small enough for people to easily commute to physical stores.

Understanding the shopping habits of people in Hong Kong, ecommerce platform Yoho

(<https://www.yohohongkong.com/>) offers customers a seamless online-to-offline (O2O) shopping experience where people can buy what they want, where they want to. When customers shop on Yoho, they can choose to purchase from its website or visit two of its easily accessible retail stores. Besides in-store pickups, Yoho offers home delivery or self-collection options from more than 2,000 locations through smart lockers and convenience stores.

“System  
reliability on  
Google Cloud is  
vital for our  
business.  
Ecommerce

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—Franz Wu, co-founder  
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Today, Yoho has more than 400,000 Hong Kong consumers on its ecommerce platform, powered by Google Cloud (<https://cloud.google.com/>). The company first deployed its website using Compute Engine (/compute) in 2017 and in 2020 migrated its legacy database from a local server to Cloud SQL (/sql).

“System reliability on Google Cloud is vital for our business. Ecommerce makes up 80% of our business, so our website always needs to be available to our customers,” says Franz Wu, co-founder and CEO at Yoho. “Modernizing our backend on Cloud SQL for scalability allows customers to enjoy uninterrupted shopping on our website at any time.”

## **Supporting high website traffic during campaign periods**

Yoho depends on ecommerce campaigns such as anniversary celebrations, 11.11, and Black Friday sales events to attract online shoppers. But before moving to Cloud SQL, Yoho’s website lacked the capacity to deliver a smooth shopping experience, particularly during successful campaigns when traffic would spike on its website.

This is because Yoho’s website was originally designed with a single-server architecture whereby the application on Compute Engine and the data on its on-premises server was tightly coupled. Product data and stock levels reflected on the website would be directly pulled from the product catalog and inventory management system in the back end. Without the ability to scale vertically, the high traffic from large-scale events would overwhelm the single database server.

Once, Yoho launched a successful campaign that attracted more than 100,000 users on the website at the same time. Unfortunately, the database shut down because it could not process the large number

of requests flooding in. This created a chain effect and caused the entire Yoho website to experience downtime, to shoppers' disappointment. "Our tech team worked overnight to troubleshoot and resolve the issue, but we lost a lot of customers that day due to the downtime," explains Franz. "After this incident, we knew we could not afford to let this happen again."

Yoho redesigned its architecture to decouple the front and back end with a queue server on Compute Engine and migrated its legacy database to Cloud SQL in May 2020. This ensures that requests exceeding the system load would be added to the queue server where the visitors wait in line for their turn, as if they were at a physical store. They would be moved up the queue and only given access to the main server to begin shopping when there was enough capacity. At the same time, the scalability of Cloud SQL enables Yoho to process a large number of transactions at any given time.

"Thanks to Google Cloud, we can now support any amount of customers on our website without worrying that it would go down," says Franz. "We chose Cloud SQL for our database because of its integration with other Google Cloud products that we use such as Compute Engine and Google Analytics. The technical support that Google Cloud offers from a local team in Hong Kong is also beneficial because we can reach out to them easily if we need help."

Yoho's move to Cloud SQL was a timely one since the company focuses primarily on selling electronic

products such as laptops and gaming consoles, which were high in demand during the stay-home notice due to COVID-19. With the reliability of Compute Engine and scalability of Cloud SQL, Yoho was well equipped to deliver a smooth user experience on its website, even when traffic increased fivefold.



## Improving operational efficiency with the power of data

Apart from its ability to scale with Cloud SQL, it's critical that Yoho delivers the right items to customers in a timely manner. With 40,000 monthly online orders, Yoho built a barcode-scanning system to track inventory and monitor sales. Using Cloud SQL as its database, the application integrates its warehouse inventory system with its point-of-sale management using Compute Engine and runs on more than 20 handheld scanners in its warehouse and retail stores. This enables customers to access real-time updates on product availability.

When Yoho receives a home delivery request, the warehouse worker scans the product barcode on the sales order and is able to pinpoint the item's exact location. Each warehouse worker can process 3,000 items a day, which is a 300% increase in efficiency compared to manual order picking, when workers had to physically go down to the right aisles in the warehouse to see if the items were available.

Pricing information is also displayed on the point-of-sale (POS) when the cashier scans the barcode, so the customer is charged correctly. In the past, when the server was down, retail staff had to take orders on pen and paper until the IT team could resolve the issue. Server outage is no longer an occurrence once Yoho migrated to Cloud SQL.

“Using a fully managed Cloud SQL database, we can focus on developing new applications such as our handheld barcode scanning software instead of worrying about maintenance,” says Franz, adding that the real-time data visibility has improved customer service.

## **Analyzing customer data and behavior on a unified platform securely**

Since moving to Cloud SQL, Yoho stores more than 100 datasets, from product browsing history to transaction data. Yoho uses Cloud SQL to analyze online data such as web-browsing behavior and offline data such as offline transaction history to understand its customers better.



Ecommerce metrics such as conversion rate help Yoho's purchasing team uncover insights about customer behavior. If the product's conversion rate is low, it is likely overpriced, so the team reprices the product to attract more shoppers and boost sales.

From product browsing history to transaction data, Yoho is able to analyze the way users interact on its platform to understand its customers better using Cloud SQL. But more than user behavior, Yoho also understands the importance of security and data privacy.

With credit cards being a common mode of payment, there's an increasing risk of chargeback from customers who dispute credit card payments and request their bank to reverse the charges, after receiving their purchase. In such cases, Yoho not only loses money but the goods as well. By analyzing offline and online transaction history, it noticed that customers who pay with cash at the retail stores are less likely to initiate a chargeback when they pay with a credit card online and, therefore, are more creditworthy. The company recognizes these cash-paying customers by their ID stored on Cloud SQL, so they can offer a wider selection of products to them the next time they log on.

Being an online store, Yoho receives a lot of confidential information from customers such as their product browsing history and credit card numbers. To protect customers, it uses security features and services such as out-of-the-box encryption and Cloud Identity and Access

Management (/iam) to keep its customer and business information safe.

## **AI transforming retail operations**

Being in the ecommerce space, Yoho understands the importance of meeting the right price point for customers. With more than 20,000 products on its website, it uses the artificial intelligence (AI) hosted on Compute Engine and analysis capability of Cloud SQL to ensure its products are always priced competitively.

Yoho created a bot that monitors market prices daily and analyzes them on Cloud SQL. The bot dynamically matches the lowest competitor price and updates the pricing between the website and the physical stores. Yoho also uses electronic shelf labels at its retail stores so that salespeople don't need to waste time replacing paper labels by hand whenever the price changes.

Just like the handheld scanning application, Yoho's software for electronic shelf labels are hosted on Compute Engine for reliability. Prices on the digital displays then sync with the pricing catalog on Cloud SQL. Besides online or offline promotions, prices are aligned across Yoho's channels to ensure customers get the best deal. Automating this process has been a timesaver for the company as the manual monitoring could take a person days to complete, and the price data may not be relevant by the time it's submitted.

“People tend to read reviews to decide what product to buy, but most customers leave reviews in Chinese. By using Cloud Translation to translate the web pages and reviews, we are able to increase the engagement of our English-speaking customers. As a result, [they] are more likely to buy from the website.”

—Franz Wu, co-founder  
and CEO, Yoho

## A multilingual platform to reach a larger market share

Although most Hong Kong customers read traditional Chinese, a small percentage of Yoho's customers are English speakers. This is why its website needs to be available in both English and Chinese languages. Instead of hiring a full-time translator to manually translate every page on its website, Yoho integrates Cloud Translation (/translate) into its website for text translation from traditional Chinese to English.

Cloud Translation automatically detects the user's preferred language based on the browser settings to display the website in traditional Chinese or English. Being able to easily navigate the platform in a familiar language also increases the chance of customers coming back and making a purchase.

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—Franz Wu, co-founder  
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# Creating a user-friendly shopping experience for retail growth

Moving forward, Yoho is exploring the use of a conversational chatbot on [Dialogflow](https://dialogflow.com/) (/dialogflow), which is part of Contact Center AI, that runs on Google Cloud to improve customer interaction online. Chatbots give customers a self-service option if they don't want to speak to a live agent over the phone. Apart from answering frequently asked questions such as store hours and delivery times, a personalized chatbot can recognize customers and their shopping preferences.

"Google Cloud gives us confidence that our ecommerce platform can support and not hamper marketing campaigns and flash sales," says Franz. "We want to further leverage Google Cloud technologies such as Dialogflow that can help us to engage customers and take our brand to the next level."



